

Assignment - II

*1

→ SAP-1 refers to simple as possible-1. It is a basic design model of a microprocessor explained by Albert Paul Malvino. This design contains all basic necessities for a functional Microprocessor. Its primary purpose is to develop a basic understanding of how a microprocessor works.

Below is the Block diagram of SAP-1 architecture.

W Bus

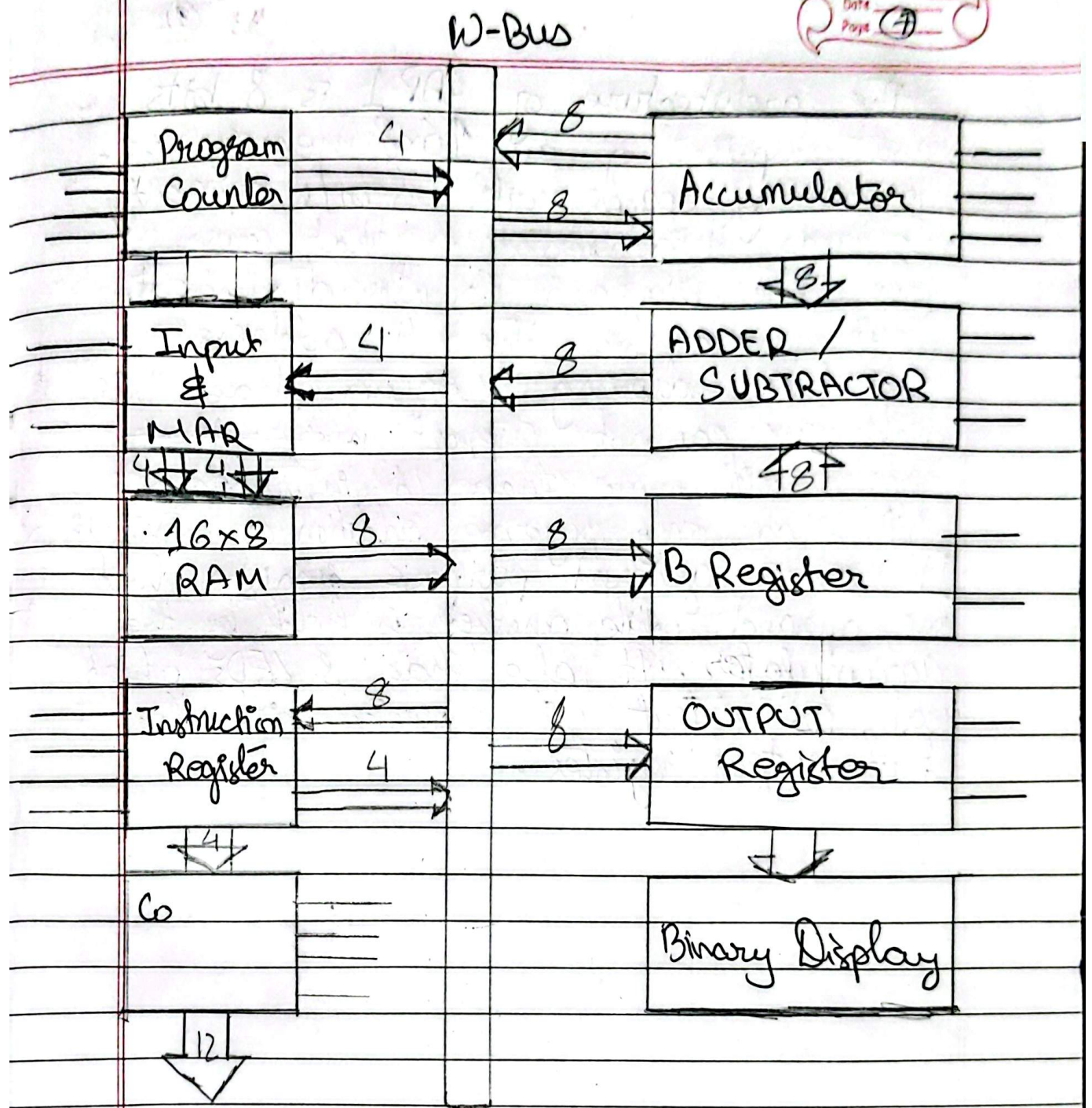


Fig: SAP-1 Architecture of a Microprocessor

The architecture of SAP-1 is 8 bits and comprises of an 16×8 memory i.e. 16 memory spaces with 8-bits capacity. It needs 4 address lines that come from the Program Counter during the run phase or the 4 address switcher during program phase. This architecture can not support ~~more than~~ program with more than 16 instructions.

SAP-1 can only perform addition or subtraction. It has a general purpose register hold one operand while another is held by the accumulator. It also has 8 LEDs which work as output unit connected with the 8 bit output register.

Q2

ACK →

Hexadecimal
Keyboard Encoder

Ready

Input port 1

8

8

Accumulator

Arbitration

Input port 2

8

8

ALU

8

Flags

PC

16

8

Temp

MAR

16

8

B

16

64K

Memory

8

8

C

MDR

8

8

Output port

Hex Display

IR

8

8

Output port

0 Serial out
7 Acknowledge

Controller
Sequences

SAP-2 is an improved computer model that accepts input through a hexadecimal keyboard encoder and processes data using registers such as the PC, MAR, MDR and IR. Instructions and data are stored in memory, and execution is controlled by a controller sequence that generates timing and control signals. Arithmetic and logical operations are performed by the ALU with results stored in accumulator and supported by temporary registers and flags. The final output is displayed through output ports on a hexadecimal display.

Q2) Compare SAP-1 and SAP-2 Architecture in different aspects.

Aspect	SAP-1	SAP-2
Complexity	Simple Architecture	Advanced Architecture.
Memory Size	16 x 8 memory (16 bytes)	64KB memory.
Program Counter.	4-bit PC	16-bit PC
Addressing Capacity	16 memory locations	65,536 memory locations.
Instruction Set.	5 basic instruction.	42 instructions.
Input ports.	No dedicated input port.	Two ports.
Output ports	One output register with LEDs	Two ports.
Display	Binary LED Display	Hexadecimal display
Arithmetic Operation.	Add and Subtract.	Multiple operations.

Logical operations.	Not supported.	Supported (AND, OR, XOR, etc.).
Flags	No flags.	Sign Flag and Zero Flag.
Jump instructions.	Not available.	Available (JMP, JN, JZ, JNZ)
Practical Capability	Basic learning model.	More realistic computer model.
Purpose	Understanding basic computer organization.	Understanding advanced microprocessors concepts.